TRIANGLE

त्रिभुज

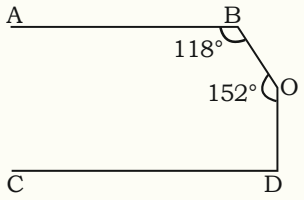
PRO LEVEL SHEET

TRIANGLE/त्रिभुज

Pro Level Sheet

1. AB||CD. ∠ABO = 118°, ∠BOD = 152°, find ∠ODC:

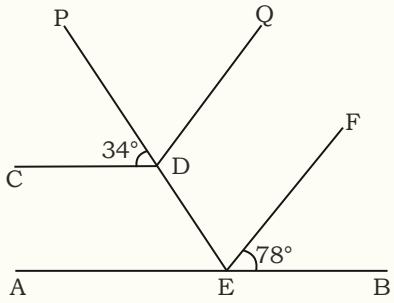
AB||CD. ∠ABO = 118°, ∠BOD = 152°, ∠ODC ज्ञात कीजिए।



(a) 70° (b) 80°
(c) 34° (d) 90°

2. In the given figure AB|| CD and EF || DQ. Find the value of ∠DEF:

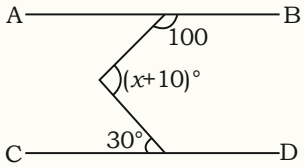
दी गई आकृति में AB||CD और EF || DQ है। ∠DEF का मान ज्ञात कीजिए:



(a) 68° (b) 78°
(c) 34° (d) 39°

3. AB|| CD, shown in the figure. Find the value of x:

AB|| CD, आकृति में दर्शाया गया है। x का मान ज्ञात कीजिए



(a) 100° (b) 90°
(c) 110° (d) 140°

4. Two parallel lines AB and CD are intersected by a transversal EF at M and N respectively.

The lines MP and NP are the bisectors of interior angle ∠BMN and ∠DNM on the same side of the transversal Then ∠MPN is equal to:

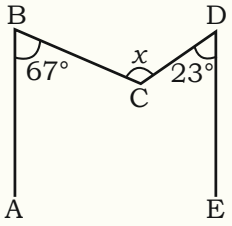
दो समांतर रेखाएँ AB और CD एक तिर्यक रेखा EF द्वारा क्रमशः M और N पर प्रतिच्छेदित होती हैं।

रेखाएँ MP और NP, तिर्यक रेखा के एक ही ओर स्थित आंतरिक कोणों ∠BMN और ∠DNM के समद्विभाजक हैं। तब ∠MPN बराबर है:

(a) 60° (b) 90°
(c) 45° (d) 120°

5. In the adjoining figure AB || DE, ∠ABC = 67° and ∠EDC = 23°. Find ∠BCD:

संलग्न आकृति AB || DE में, ∠ABC = 67° और ∠EDC = 23° है। ∠BCD ज्ञात कीजिए:



(a) 90° (b) 44°
(c) 46° (d) none of the above

6. The measure of sides of a triangle is $(x^2 - 1)$, $(x^2 + 1)$ and $2x$ then find the triangle.

किसी त्रिभुज की भुजा $(x^2 - 1)$, $(x^2 + 1)$, $2x$ है त्रिभुज होगा–

(a) Equilateral
(b) Isosceles
(c) Right angled triangle
(d) Acute angled triangle

7. Let ABC be an equilateral triangle and AD perpendicular to BC then $AB^2 + BC^2 + CA^2$ = ?

मान लीजिए ABC एक समबाहु त्रिभुज है और AD, BC पर लंबवत है तो $AB^2 + BC^2 + CA^2$ = ?

(a) $2AD^2$ (b) $3AD^3$
(c) $4AD^2$ (d) $5AD^3$

8. If AD, BE, CF are the medians of a ΔABC then the correct relation between the sum of the squares of sides to the sum of the squares of medians is.

यदि AD, BE, CF एक ΔABC की माध्यिकाएँ हैं तो भुजाओं के वर्गों के योग का माध्यिका के वर्गों के योग से सही संबंध है।

(a) $2(AB^2 + BC^2 + AC^2) = 3(AD^2 + BE^2 + CF^2)$
(b) $4(AB^2 + BC^2 + AC^2) = 3(AD^2 + BE^2 + CF^2)$
(c) $3(AB^2 + BC^2 + AC^2) = 4(AD^2 + BE^2 + CF^2)$
(d) None of the above

9. PQR is an isosceles triangle such that PQ = QR = 10 cm and ∠PQR = 90°. What is the length of the perpendicular drawn from Q on PR?

PQR एक ऐसा समद्विबाहु त्रिभुज है कि PQ = QR = 10 cm और ∠PQR = 90° है। भुजा PR पर बिंदु Q से खींचे जाने वाले लंब की लंबाई बताइए।

(a) $4\sqrt{2}$ cm (b) $7\sqrt{2}$ cm
(c) $6\sqrt{2}$ cm (d) $5\sqrt{2}$ cm

10. In triangle ABC, a line is drawn from the vertex A to a point D on BC. If BC = 9 cm and DC = 3 cm, then what is the ratio of the areas of triangle ABD and triangle ADC respectively?

त्रिभुज ABC में, BC पर बिंदु A से बिंदु D तक रेखा खींची जाती है। यदि BC = 9 cm और DC = 3 cm है, तो क्रमशः त्रिभुज ABD और त्रिभुज ADC के क्षेत्रफलों का अनुपात क्या है?

(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 4 : 1

11. ΔABC and ΔDEF are two triangles such that $\Delta ABC \cong \Delta FDE$. If AB = 5 cm, $\angle B = 40°$ and $\angle A = 80°$, then which of the following option is true?

ΔABC और ΔDEF दो ऐसे त्रिभुज हैं कि $\Delta ABC \cong \Delta FDE$ है। यदि AB = 5 cm, $\angle B = 40°$ और $\angle A = 80°$ है, तो निम्नलिखित में से कौन सा विकल्प सत्य है?

(a) DE = 5 cm, $\angle E = 60°$
(b) DE = 5 cm, $\angle F = 60°$
(c) DE = 5 cm, $\angle D = 60°$
(d) DF = 5 cm, $\angle E = 60°$

12. If ΔABC is an obtuse-angled triangle, the ratio of the angles $\angle A$, $\angle B$ and $\angle C$ can be:

यदि ΔABC एक अधिककोण त्रिभुज है, तो कोण $\angle A$, $\angle B$ और $\angle C$ का अनुपात हो सकता है:

(a) 4 : 9 : 3
(b) 11 : 14 : 21
(c) 1 : 2 : 3
(d) 5 : 8 : 12

13. In ΔABC, $\angle C = 90°$ and D is a point on CB such that AD is the bisector of $\angle A$. If AC = 5 cm and BC = 12 cm, then what is the length of AD?

ΔABC में, $\angle C = 90°$ है और D, CB पर एक ऐसा बिंदु है जिससे कि AD, $\angle A$ का समद्विभाजक है। यदि AC = 5 सेमी और BC = 12 सेमी है, तो AD की लम्बाई कितनी है?

(a) $\frac{10}{3}$ cm (b) $\frac{5\sqrt{3}}{6}$ cm
(c) $\frac{5\sqrt{13}}{3}$ cm (d) $\frac{20}{3}$ cm

14. The sides of a scalene acute angle triangle are 14,22 and *x*. *x* is an integer and 22 is the largest side of triangle, what will be the average of minimum and maximum value of *x*?

एक न्यून कोण विषमबाहु त्रिभुज की भुजाएँ 14, 22 और *x* हैं। *x* एक पूर्णांक है जबकि त्रिभुज की सबसे बड़ी भुजा 22 है। *x* के न्युनतम और महत्तम मान का औसत क्या होगा?

(a) 18 (b) 19
(c) 21 (d) 20.5

15. If two sides of an obtuse angle triangle are 8 cm and 15 cm respectively. How many triangles are possible given the length of the third side is an integer (in cms)?

यदि एक अधिक कोण त्रिभुज की दो भुजाएँ क्रमशः 8 सेमी और 15 सेमी हैं। यदि तीसरी भुजा की लंबाई एक पूर्णांक (सेमी में) है तो कितने त्रिभुज संभव हैं?

(a) 6 (b) 10
(c) 9 (d) None

16. In a triangle ABC, AC = 20 cm, BC = 10 cm and area of triangle is 80 sq cm then find the length of AB.

एक त्रिभुज ABC में, AC = 20 सेमी, BC = 10 सेमी और त्रिभुज का क्षेत्रफल 80 वर्ग सेमी है, तो AB की लंबाई ज्ञात कीजिए।

(a) $3\sqrt{39}$ cm (b) $2\sqrt{78}$ cm
(c) $2\sqrt{52}$ cm (d) $2\sqrt{65}$ cm

17. ABC is an equilateral triangle and CD is the internal bisector of $\angle C$. If DC is produced to E such that AC = CE, then $\angle CAE$ is equal to:

ABC एक समबाहु त्रिभुज है और CD, $\angle C$ का आंतरिक समद्विभाजक है। यदि DC को E तक इस प्रकार बढ़ाया जाए कि AC = CE है, तो $\angle CAE$ बराबर है:

(a) 45° (b) 75°
(c) 30° (d) 15°

Direction(1-2):- Consider the following for the next two (02) questions:

अगले दो (02) प्रश्नों के लिए निम्नलिखित पर विचार कीजिए:

In ΔABC, a, b and c are the lengths of the sides and p, q and r are the lengths of its medians.

एक ΔABC में, a, b और c भुजाओं की लम्बाई हैं और p, q और r इसकी माध्यिकाओं की लम्बाई हैं।

18. Which one of the following is correct?

निम्नलिखित में से कौन-सा सही है?

(a) $2(p + q + r) = (a + b + c)$
(b) $2(p + q + r) > 3(a + b + c)$
(c) $2(p + q + r) < 3(a + b + c)$
(d) $11(p + q + r) > 10(a + b + c)$

19. Which one of the following is correct?

निम्नलिखित में से कौन-सा सही है?

(a) $(a + b + c) > (p + q + r)$
(b) $3(a + b + c) < 4(p + q + r)$
(c) $2(a + b + c) > 3(p + q + r)$
(d) $3(a + b + c) > 4(p + q + r)$

20. Consider the following statements/निम्नलिखित कथनों पर विचार कीजिए:

1. The orthocentre of a triangle always lies inside the triangle/एक त्रिभुज का लंब केन्द्र हमेशा उस त्रिभुज के अंदर ही होता है।
2. The centroid of a triangle always lies inside the triangle/एक त्रिभुज का केन्द्रक हमेशा उस त्रिभुज के अंदर ही होता है।
3. The orthocentre of a right angled triangle lies on the triangle/एक समकोण त्रिभुज का लंब केन्द्र उस त्रिभुज पर स्थित होता है।
4. The centroid of a right angled triangle lies on the triangle/एक समकोण त्रिभुज का केन्द्रक उस त्रिभुज पर स्थित होता है।

Which of the above statements are correct/उपर्युक्त में से कौन-से कथन सही हैं?

(a) 1 and 2 (b) 1 and 4

(c) 2 and 3 (d) 2 and 4

21. ABC is a triangle right angled at C with BC = a and AC = b. If p is the length of the perpendicular from C on AB, then which one of the following is correct?

ABC एक त्रिभुज है जो C पर समकोण है, जहाँ BC = a और AC = b है। यदि C से AB पर खींचे जाने वाले लंब की लंबाई p है, तो निम्नलिखित में से कौन-सा सही है?

(a) $a^2b^2 = p^2(a^2 + b^2)$

(b) $a^2b^2 = p^2(a^2 - a^2)$

(c) $2a^2b^2 = p^2(a^2 + b^2)$

(d) $a^2b^2 = 2p^2(a^2 + b^2)$

22. In ΔABC, the sides AB, AC are produced and the bisectors of exterior angles of ∠ABC and ∠ACB intersect at D. If ∠BAC = 50° then ∠BDC is equal to.

एक ΔABC में, भुजाएँ AB, AC बढ़ाई जाती हैं और ∠ABC और ∠ACB के बाह्य कोणों के समद्विभाजक D पर प्रतिच्छेद करते हैं। यदि ∠BAC = 50° है तो ∠BDC बराबर है।

(a) 115° (b) 65°

(c) 55° (d) 40°

23. AD is the median of the ΔABC. If P is any point on AD, then which one of the following is correct?

ΔABC की माध्यिका AD है। यदि AD पर कोई बिन्दु P है, तो निम्नलिखित में से कौन-सा एक सही है?

(a) Area of ΔPAB is greater than the area of ΔPAC

ΔPAB का क्षेत्रफल, ΔPAC के क्षेत्रफल से अधिक है।

(b) Area of ΔPAB is equal to area of ΔPAC

ΔPAB का क्षेत्रफल, ΔPAC के क्षेत्रफल के बराबर है।

(c) Area of ΔPAB is one-fourth of the area of ΔPAC

ΔPAB का क्षेत्रफल, ΔPAC के क्षेत्रफल का एक-चौथाई है।

(d) Area of ΔPAB is half of the area of ΔPAC

ΔPAB का क्षेत्रफल, ΔPAC के क्षेत्रफल का आधा है।

24. ΔABC is a right angled triangle at C. Let p be the length of the perpendicular drawn from C on AB. If BC = 6 cm and CA = 8 cm, then what is the value of p?

ΔABC एक समकोण त्रिभुज है जिसका कोण C समकोण है। मान लीजिए C से AB पर खींचे गए लम्ब की लंबाई p है। यदि BC = 6 सेमी और CA = 8 सेमी है, तो p का मान क्या है?

(a) 5.4 cm

(b) 5 cm

(c) 4.8 cm

(d) 4.2 cm

25. In ΔABC, AD is the bisector of ∠A. If AB = 5 cm, AC = 7.5 cm and BC = 10 cm, then what is the distance of D from the mid-point of BC (in cm) ?

ΔABC में, AD, ∠A का समद्विभाजक है। यदि AB = 5 सेमी, AC = 7.5 सेमी और BC = 10 सेमी है, तो BC के मध्य बिन्दु से D की दूरी (सेमी में) ज्ञात करें।

(a) 2 (b) 1.5

(c) 2.2 (d) 1

26. ABC is a right-angled triangle, right angled at A in such a way that AB = AC and the bisector of angle B intersects the side AC at D. DM is perpendicular to BC, intersecting BC at M. If AB = 12 cm and BC = $12\sqrt{2}$ cm, then find the length of side AD.

ABC एक समकोण त्रिभुज है, जो A पर इस प्रकार समकोण है कि AB = AC तथा कोण B का समद्विभाजक भुजा AC को D पर प्रतिच्छेद करता हैं। DM, BC के लंबवत है, जो M पर BC को प्रतिच्छेद करता है। यदि AB = 12 सेमी तथा BC = $12\sqrt{2}$ सेमी है, तो भुजा AD की लंबाई ज्ञात कीजिए।

(a) $12(\sqrt{2} + 2)$cm

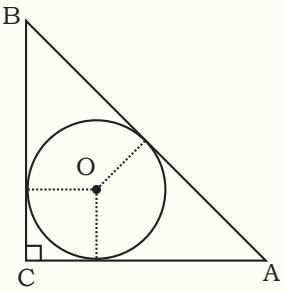
(b) $12(\sqrt{2} - 2)$cm

(c) $12(\sqrt{2} - 1)$cm

(d) $12(\sqrt{2} + 1)$cm

27. A circle of radius 4 cm is drawn inscribed in a right angle triangle ΔABC, right angled at C. If AC =12 cm, then the value of CB is:

एक समकोण त्रिभुज ΔABC, जिसमें कोण C समकोण है, उसमें 4 सेमी त्रिज्या वाला एक वृत्त खींचा जाता है। यदि AC = 12 सेमी है, तो CB का मान है:



(a) 8 cm (b) 12 cm

(c) 20 cm (d) 16 cm

28. In ΔMNO, XA, XB and XC are perpendicular bisectors on sides MN, ON and OM respectively intersect each other at X. If ∠NMO = 55°, then what is the value of ∠NXO?

O में XA, XB और XC क्रमशः MN, ON और OM भुजाओं के लंब समद्विभाजक है। जो परस्पर X पर प्रतिच्छेद करते हैं। यदि ∠NMO = 55° है, तो ∠NXO का मान क्या है?

(a) 130° (b) 110°

(c) 95° (d) 125°

29. The length of a side of an equilateral triangle is 18 cm. The area (in cm^2) of the region lying between the circumcircle and the incircle of the triangle is

[use $\pi = \frac{22}{7}$]

एक समबाहु त्रिभुज की भुजा की लंबाई 18 सेमी है। त्रिभुज के परिवृत्त और अंतः वृत्त के बीच स्थित क्षेत्र का क्षेत्रफल ($सेमी^2$ में) कितना है?

[$\pi = \frac{22}{7}$ का उपयोग कीजिए]

(a) $254\frac{3}{7}$ (b) $254\frac{1}{7}$

(c) $254\frac{2}{7}$ (d) $254\frac{4}{7}$

30. ABC is an isosceles right angle triangle. $\angle ABC = 90°$ and AB = 12 cm. What is the ratio of the radius of the circle inscribed in it to the radius of the circle circumscribing ΔABC?

ABC एक समद्विबाहु समकोण त्रिभुज है। $\angle ABC = 90°$ तथा AB = 12 सेमी है। ΔABC के अंतर्गत वृत्त की त्रिज्या का उसके परिगत वृत्त की त्रिज्या से अनुपात क्या है?

(a) $6 - 3\sqrt{2} : 6\sqrt{2}$ (b) $2 - \sqrt{2} : \sqrt{2}$

(c) $6 - \sqrt{2} : 3\sqrt{2}$ (d) $6 - 3\sqrt{2} : \sqrt{2}$

31. An isosceles ΔMNP is inscribed in a circle. If $MN = MP = 16\sqrt{5}$ cm, and NP = 32 cm. What is the radius (in cm) of the circle?

एक समद्विबाहु ΔMNP एक वृत्त में बना है। यदि $MN = MP = 16\sqrt{5}$ सेमी और NP = 32 सेमी है तो वृत्त की त्रिज्या की लंबाई (सेमी में) कितनी है?

(a) $20\sqrt{5}$ (b) 18

(c) $18\sqrt{5}$ (d) 20

32. ΔXYZ is an isosceles triangle such that XY = XZ. It is given that $YS \perp XZ$ and $ZT \perp XY$. What is the relation between YS and ZT?

ΔXYZ एक समद्विबाहु त्रिभुज है जिसमें XY = XZ है। यह दिया गया है कि 'YS $\perp$ XZ और ZT $\perp$ XY है। YS और ZT के बीच क्या संबंध है?

(a) YS = ZT (b) YS = 2 ZT

(c) YS < ZT (d) YS > ZT

33. In ΔABC, AB = 6 cm, AC = 8 cm, and BC = 9 cm. The length of median AD is:

ΔABC में, AB = 6 सेमी, AC = 8 सेमी और BC = 9 सेमी है। माध्यिका AD की लंबाई है:

(a) $\sqrt{\frac{115}{2}}$ cm (b) $\sqrt{\frac{317}{2}}$ cm

(c) $\sqrt{\frac{313}{2}}$ cm (d) $\frac{\sqrt{119}}{2}$ cm

34. In ΔSTU, SE and TF are medians intersect each other at O. If UT = 34 cm, TF = 22.5 cm and SE = 24 cm, then what is the value of $\angle TOE$?

ΔSTU में SE और TF माध्यिकाएँ एक-दूसरे को O पर प्रतिच्छेद करती हैं। यदि UT = 34 सेमी, TF = 22.5 सेमी और SE = 24 सेमी है, तो $\angle TOE$ का मान क्या है?

(a) 75° (b) 60 °

(c) 45° (d) 90°

35. ΔABC is a right angled at A and AD is the altitude to BC. If AB = 7 cm and AC = 24 cm. Find the ratio of AD is to AM if M is the mid-point of BC :

ΔABC एक समकोण त्रिभुज है जिसका A समकोण है और AD BC पर शीर्षलंब है। यदि AB = 7 सेमी और AC = 24 सेमी है। यदि M, BC का मध्य-बिंदु है, तो AD का AM से अनुपात ज्ञात कीजिए:

(a) 25 : 41

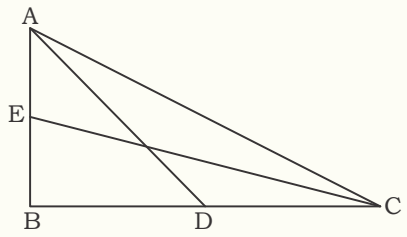
(b) 32 : 41

(c) 336 : 625

(d) 625 : 336

36. In figure, ABC is a right triangle, right angled at B. AD and CE are the two medians drawn from A and C respectively. If AC = 5 cm and AD = $\frac{3\sqrt{5}}{2}$ cm, find the length of CE:

आकृति में, ABC एक समकोण त्रिभुज है, जिसका कोण B समकोण है। AD और CE क्रमशः A और C से खींची गई दो माध्यिकाएँ हैं। यदि AC = 5 सेमी और AD = $\frac{3\sqrt{5}}{2}$ सेमी है, तो CE की लंबाई ज्ञात कीजिए।

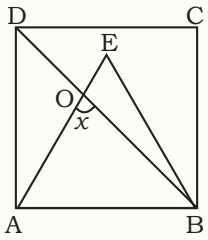


(a) $2\sqrt{5}$ cm (b) 2.5 cm

(c) 5 cm (d) $4\sqrt{2}$ cm

37. In the figure ΔABE is an equilateral triangle in a square ABCD. Find the value of angle x in degrees:

आकृति में ΔABE एक वर्ग ABCD में स्थित एक समबाहु त्रिभुज है। कोण x का मान अंशों में ज्ञात कीजिए।



(a) 60° (b) 45°

(c) 75° (d) 90°

38. The hypotenuse of a right triangle is 2 more than twice one of the sides, while the third side is 13 more than half the hypotenuse. Find the length of the median drawn on the hypotenuse.

एक समकोण त्रिभुज का कर्ण, किसी एक भुजा के दोगुने से 2 अधिक है, जबकि तीसरी भुजा कर्ण के आधे से 13 अधिक है। कर्ण पर खींची गई माध्यिका की लंबाई ज्ञात कीजिए।

(a) 15 (b) 18

(c) 17 (d) 20

39. The lengths of three medians of a triangle are 3 cm, 6 cm and 5 cm. What is the area of the triangle?

एक त्रिभुज की तीन माध्यिकाओं की लंबाई 3 सेमी, 6 सेमी और 5 सेमी है। त्रिभुज का क्षेत्रफल कितना है?

(a) $\frac{\sqrt{405}}{3}$ cm^2 (b) $\frac{\sqrt{345}}{3}$ cm^2

(c) $\frac{4}{3}\sqrt{56}$ cm^2 (d) $\frac{\sqrt{255}}{3}$ cm^2

40. In ΔPQR, PQ = PR and S is a point on QR such that $\angle PSQ = 96° + \angle QPS$ and $\angle QPR = 132°$. What is the measure of $\angle PSR$?

ΔPQR में, PQ = PR और S रेखा QR पर इस प्रकार स्थित एक बिंदु है कि $\angle PSQ = 96° + \angle QPS$ और $\angle QPR = 132°$ है। $\angle PSR$ का माप बताइए।

(a) 45° (b) 56°
(c) 54° (d) 52°

41. In a ΔABC, points P, Q and R are taken on AB, BC and CA, respectively, such that BQ = PQ and QC = QR. If ∠BAC = 75°, what is the measure of ∠PQR (in degrees)?

ΔABC में, बिंदु P, Q और R क्रमशः AB, BC और CA पर इस प्रकार लिए गए हैं कि BQ = PQ और QC = QR है। यदि ∠BAC = 75° है, तो ∠PQR का माप (डिग्री में) क्या है?

(a) 40° (b) 30°
(c) 50° (d) 75°

42. In ΔPQR, S is a point on the side QR such that ∠QPS = $\frac{1}{2}$∠PSR, ∠QPR = 78°and ∠PRS = 44°. What is the measure of ∠PSQ?

ΔPQR में, S एक बिंदु है, जो भुजा QR पर इस प्रकार स्थित है कि ∠QPS = $\frac{1}{2}$∠PSR, ∠QPR = 78° और ∠PRS = 44° है। ∠PSQ की माप क्या है?

(a) 68° (b) 64°
(c) 58° (d) 56°

43. Select the correct statement about the properties of a triangle.

त्रिभुज के गुणधर्मों के बारे में सही कथन का चयन कीजिये।

(a) The sum of two sides may be equal to the third side.
दो भुजाओं का योग तीसरी भुजा के बराबर हो सकता है।

(b) The sum of two sides is always equal to the third side.
दो भुजाओं का योग सदैव तीसरी भुजा के बराबर होता है।

(c) The sum of two sides is always greater than the third side
दो भुजाओं का योग सदैव तीसरी भुजा से अधिक होता है।

(d) The sum of two sides is always less than the third side.
दो भुजाओं का योग तीसरी भुजा से छोटा होता है।

44. If the sides of a triangle are 7, 12 and x, and x is an integer, then find the number of possible values of x.

यदि किसी त्रिभुज की भुजाएँ 7, 12 और x हैं, और x एक पूर्णांक है, तो x के संभावित मानों की संख्या ज्ञात कीजिए।

(a) 14 (b) 13
(c) 15 (d) 12

45. If in ΔPQR, ∠P = 120°, PS⊥QR at S and PQ + QS = SR, then the measure of ∠Q is-

ΔPQR में, ∠P = 120°, S पर PS⊥QR और PQ + QS = SR है, तो ∠Q की माप क्या होगी?

(a) 20° (b) 50°
(c) 40° (d) 30°

46. A triangle ABC, if sides AB = $6\sqrt{3}$, AC = 12 and BC = 6 then the angles B are equal to:

एक त्रिभुज ABC में, यदि भुजा AB = $6\sqrt{3}$, AC = 12 और BC = 6 है तो कोण B समान है:

(a) 120° (b) 60°
(c) 90° (d) 45°

47. In a quadrilateral ABCD, AB = 6 cm, BC = 18 cm, CD = 6 cm and DA = 10 cm. If the diagonal BD = x, then which of the following is correct?

चतुर्भुज ABCD में, AB = 6 सेमी, BC = 18 सेमी, CD = 6 सेमी और DA = 10 सेमी है। यदि विकर्ण BD = x है, तो निम्नलिखित में से कौन सा सही है?

(a) $8 < x < 12$ (b) $12 < x < 16$
(c) $16 < x < 18$ (d) $18 < x < 20$

48. The lengths of sides of a triangle are $3x$, $4\sqrt{y}$, $5\sqrt[3]{z}$, where $3x < 4\sqrt{y} < 5\sqrt[3]{z}$. If one of the angles is 90°, then what are the minimum integer values of x, y, z respectively?

एक त्रिभुज की भुजाओं की लंबाइयाँ $3x$, $4\sqrt{y}$, $5\sqrt[3]{z}$ हैं, जहाँ $3x < 4\sqrt{y} < 5\sqrt[3]{z}$ है। यदि त्रिभुज का एक कोण 90° है, तो x, y, z के न्यूनतम पूर्णांक मान क्रमशः क्या हैं?

(a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 1, 1 (d) 3, 4, 5

49. In a ΔABC, if 2∠A = 3∠B = 6∠C, then what is ∠A + ∠C equal to?

एक ΔABC में, यदि 2∠A = 3∠B = 6∠C है, तो ∠A + ∠C किसके बराबर है?

(a) 90° (b) 120°
(c) 135° (d) 150°

50. The lengths of the sides of a right-angled triangle are consecutive even integers (in cm). What is the product of these integers?

एक समकोण त्रिभुज की भुजाओं की लंबाइयाँ क्रमागत सम पूर्णांक (सेमी में) हैं। इन पूर्णांकों का गुणनफल क्या है?

(a) 60 (b) 120
(c) 360 (d) 480

51. In ΔABC, AB = AC and BC is produced to D such that ∠ACD = x, then what is ∠BAC equal to

एक ΔABC में, AB = AC और BC को D तक इस प्रकार बढ़ाया गया है कि ∠ACD = x, तो ∠BAC किसके बराबर है?

(a) $2x - 90°$ (b) $180° - 2x$
(c) $2x - 180°$ (d) $\frac{x}{2}$

52. In ΔABC, ∠A = 60°. What is $AB^2 + AC^2 - BC^2$ equal to?

एक ΔABC में, ∠A = 60°. है। $AB^2 + AC^2 - BC^2$ किसके बराबर है?

(a) AB.AC (b) AB.BC
(c) AC.BC (d) 2AB.AC

53. The perpendicular AD on the base BC of a ΔABC intersects BC at D so that DB = 3CD. Which one of the following is correct.

एक ΔABC के आधार BC पर डाला गया लम्ब AD, BC को D पर इस प्रकार प्रतिच्छेद करता है कि DB = 3 CD है। निम्नलिखित में से कौन सा एक सही है?

(a) $2(AB + AC)(AB - AC) = BC^2$
(b) $3(AB + AC)(AB - AC) = 2BC^2$
(c) $4(AB + AC)(AB - AC) = 3BC^2$
(d) $5(AB + AC)(AB - AC) = 4BC^2$

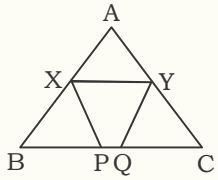
54. ABC is a triangle right angled at B. Let M and N be two points on AB such that AM = MN = NB. Let P and Q be two points on AC such that PM is parallel to QN and QN is parallel to CB. If BC = 12 cm, then what is (PM + QN) equal to?

ABC एक त्रिभुज है जिसका कोण B समकोण है। मान लीजिए M और N, AB पर दो बिंदु हैं जैसे कि AM = MN = NB मान लीजिए P और Q, AC पर दो बिंदु हैं जैसे कि PM, QN के समानांतर है और QN, CB के समानांतर है। यदि BC = 12 सेमी है, तो (PM + QN) किसके बराबर है?

(a) 10 cm (b) 11 cm
(c) 12 cm (d) 13 cm

55. In the figure given below, ABC is an equilateral triangle with each side of length 30 cm. XY is parallel to BC, XP is parallel to AC and YQ is parallel to AB. If XY + XP + YQ is 40 cm, then the value of PQ is.

नीचे दी गई आकृति में, ABC एक समबाहु त्रिभुज है, जिसकी प्रत्येक भुजा 30 सेमी लंबाई की है। XY, BC, के समान्तर है, XP, AC के समान्तर है। यदि XY + XP + YQ, 40 सेमी है, तो PQ का मान क्या है?



(a) 5 cm (b) 12 cm
(c) 15 cm (d) 10 cm

56. In a ΔPQR, ∠P = 30° and straight line UV is drawn parallel to QR, intersecting PQ at S and PR at T. Which of the following will always be true?

ΔPQR में, ∠P = 30° और सीधी रेखा UV, QR के समानांतर खींची गई है, जो PQ को S पर और PR को T पर प्रतिच्छेद करती है। निम्न में से कौन सा हमेशा सत्य होगा?

(1) $\frac{PS}{PQ} = \frac{PT}{PR}$

(2) $ST = \frac{QR}{2}$

(3) ΔPST is similar to ΔPQR.

(a) Only (1) & (3)
(b) Only (1) & (2)
(c) (1), (2) & (3)
(d) Only (1)

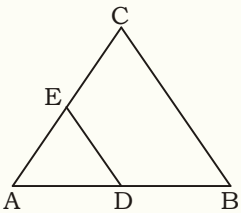
57. In ΔABC, DE || AC, where D and E are the points on sides AB and BC, respectively. If BD = 8 cm and AD = 7 cm, then what is the ratio of the area of ΔBDE to the trapezium ADEC?

ΔABC में, DE || AC है, जहाँ D और E क्रमशः भुजाओं AB और BC पर स्थित बिंदु हैं। यदि BD = 8 सेमी और AD = 7 सेमी है, तो ΔBDE के क्षेत्रफल और समलंब ADEC के क्षेत्रफल का अनुपात ज्ञात कीजिए।

(a) 161 : 64 (b) 65 : 162
(c) 64 : 161 (d) 162 : 65

58. In the given figure, DE ∥ BC. If AD = x, DB = $x - 2$, AE = $x + 2$ and EC = $x - 1$, then find the value of x.

दी गई आकृति में, DE || BC यदि AD = x, DB = $x - 2$, AE = $x + 2$ और EC = $x - 1$ हो, तो x का मान ज्ञात कीजिए।



(a) 3 (b) 4
(c) 5 (d) 2

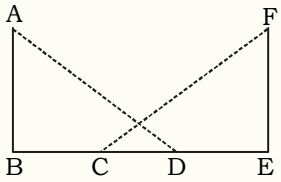
59. The midpoints of sides BC, CA, AB of ΔABC are points D, E, F respectively. If the area of ΔABC is 20 square cm, then what will be the area of trapezium FBCE?

ΔABC की भुजाओं BC, CA, AB के मध्यबिंदु क्रमशः बिंदु D, E, F हैं। यदि ΔABC का क्षेत्रफल 20 वर्ग सेमी हो, तो, समलम्ब चतुर्भुज FBCE का क्षेत्रफल क्या होगा?

(a) 10 cm^2 (b) 5 cm^2
(c) 15 cm^2 (d) 12 cm^2

60. In ΔABD and ΔFEC, ∠BAD = 60°, l(BD) = l(EC), ∠ABD = ∠FEC = 90°, and l(AB) = l(FE). Find the ratio of $\frac{\angle BAD}{\angle FCE}$.

ΔABD और ΔFEC में, ∠BAD = 60°, l(BD) = l(EC), ∠ABD = ∠FEC = 90° और l(AB) = l(FE) है। $\frac{\angle BAD}{\angle FCE}$ का अनुपात ज्ञात कीजिए।



(a) 2 : 3 (b) 1 : 2
(c) 2 : 5 (d) 2 : 1

61. In ΔABC, M is the midpoint of the side AB. N is a point in the interior of ΔABC such that CN is the bisector of ∠C and CN ⊥ NB. What is the length (in cm) of MN, if BC = 10 cm and AC = 15 cm?

ΔABC में, भुजा AB का मध्यबिंदु M है। N एक ऐसा बिंदु है जो ΔABC के अंदर इस प्रकार स्थित है, कि CN, ∠C का समद्विभाजक है और CN ⊥ NB है। यदि BC = 10 सेमी और AC = 15 सेमी, तो MN की लंबाई (सेमी में) कितनी है?

(a) 2.5 (b) 2
(c) 5 (d) 4

62. In ΔPQR, PQ = QR and O is an interior point of ΔPQR such that ∠OPR = ∠ORP.

ΔPQR में, PQ = QR है और ΔPQR में O इस प्रकार से एक अंतः बिंदु है कि ∠OPR = ∠ORP है।

Consider the following statements:
निम्नलिखित कथनों पर विचार कीजिए:

(i) ΔPOR is an isosceles triangle.
ΔPOR एक समद्विबाहु त्रिभुज है।

(ii) O is the centroid of ΔPQR.
O बिंदु ΔPQR का केन्द्रक है।

(iii) ΔPQO is congruent to ΔRQO.
ΔPQO, ΔRQO के सर्वांगसम है।

Which of the above statements are correct?
उपरोक्त कथनों में से कौन से कथन सही हैं?

(a) Only (i) and (ii)/केवल (i) और (ii)
(b) Only (i) and (iii)/केवल (i) और (iii)
(c) Only (ii) and (iii)/केवल (ii) और (iii)
(d) Only (ii)/केवल (ii)

63. In a ΔABC, DE is parallel to BC, AD = a, DB = a + 4, AE = 2a + 3, EC = 7a. What is the value of 'a' if a > 0?

एक त्रिभुज ABC में DE, BC के समानांतर है। AD = a, DB = a + 4, AE = 2a + 3, EC = 7a है। यदि a > 0 है तो a का मान क्या होगा?

(a) 3 (b) 5
(c) 6 (d) 4

64. In ΔABC, a straight line parallel to the side BC meets AB and AC at the points P and Q, respectively. If AP = QC, the length of AB is 16 cm and the length of AQ is 4 cm, then the length of (in cm) CQ is:

ΔABC में, भुजा BC के समानांतर सीधी रेखा AB और AC को बिंदु P और Q पर, परस्पर मिलती है। यदि AP = QC, है, AB की लंबाई 16 सेमी है और AQ की लंबाई 4 सेमी है, तो CQ की लंबाई (सेमी में) है:

(a) $\left(2\sqrt{21}+2\right)$ (b) $\left(2\sqrt{18}-2\right)$
(c) $\left(2\sqrt{17}-2\right)$ (d) $\left(2\sqrt{19}+2\right)$

65. In a ΔPQR, RS intersects PQ at point S. The sides of the triangle QR = 36 cm, SQ = 27 cm, RS = 18 cm and ∠QRS = ∠QPR. What is the ratio of the perimeter of ΔPRS to that of ΔQSR?

त्रिभुज PQR में, RS, PQ को बिंदु S पर प्रतिच्छेदित करती है। त्रिभुज की भुजाएं QR = 36 सेमी, SQ = 27 सेमी, RS = 18 सेमी हैं और ∠QRS = ∠QPR है। ΔPRS के परिमाप और ΔQSR के परिमाप का अनुपात कितना है?

(a) $\frac{8}{6}$ (b) $\frac{9}{12}$
(c) $\frac{7}{9}$ (d) $\frac{5}{8}$

66. In ΔABC, DE||AB, where D and E are points on sides AC and BC, respectively. F is a point between C and D such that EF || BD. If AD = 15cm, DC = 10cm, then the length of CF is:

ΔABC में, DE||AB है, जहां D और E क्रमशः भुजा AC और BC पर स्थित बिंदु हैं। C और D के बीच बिंदु F इस प्रकार है कि EF||BD है। यदि AD = 15 सेमी, DC = 10 सेमी है, तो CF की लंबाई ज्ञात करें।

(a) 3 cm (b) 7.5 cm
(c) 4 cm (d) 5 cm

67. O is a point in the interior of ΔABC such that OA = 12 cm, OC = 9 cm, ∠AOB = ∠BOC = ∠COA and ∠ABC = 60°. What is the length (in cm) of OB?

O एक बिंदु है जो ΔABC के अंदर इस प्रकार है कि OA = 12 सेमी, OC = 9 सेमी, ∠AOB = ∠BOC = ∠COA और ∠ABC = 60° है। OB की लंबाई (सेमी में) कितनी है?

(a) $6\sqrt{3}$ (b) $4\sqrt{6}$
(c) $4\sqrt{3}$ (d) $6\sqrt{2}$

68. In ΔABC, P is a point on AB such that PB : AP = 3 : 4 and PQ is parallel to AC. If AR and QS are perpendicular to PC and QS = 9 cm, what is the length (in cm) of AR?

ΔABC में, AB पर एक बिंदु P इस प्रकार है कि BP : AP = 3 : 4 है और PQ, AC के समांतर है। यदि AR और QS, PC के लंबवत हैं और QS = 9 सेमी है, तो AR की लंबाई (सेमी में) कितनी है?

(a) 28 (b) 35
(c) 21 (d) 14

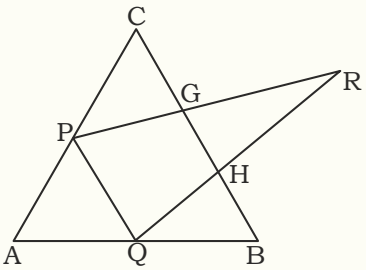
69. ΔABC is similar to ΔPQR. AB = 4 cm, BC = 8 cm and AC = 10 cm. If QR = 16 cm, then what is the area of ΔPQR?

ΔABC, ΔPQR के समरूप है। AB = 4 सेमी, BC = 8 सेमी और AC = 10 सेमी है। यदि QR = 16 सेमी है, तो ΔPQR का क्षेत्रफल कितना है?

(a) $2\sqrt{231}$ cm² (b) $4\sqrt{231}$ cm²
(c) $4\sqrt{462}$ cm² (d) $8\sqrt{462}$ cm²

70. In the given figure, P and Q are the mid-points of AC and AB. Also, PG = GR and HQ = HR. What is the ratio of area of ΔPQR : area of ΔABC?

दी गई आकृति में, P और Q क्रमशः AC और AB के मध्यबिंदु हैं। साथ ही, PG = GR और HQ = HR है। तो क्षेत्रफल (ΔPQR): क्षेत्रफल (ΔABC) का अनुपात क्या होगा?



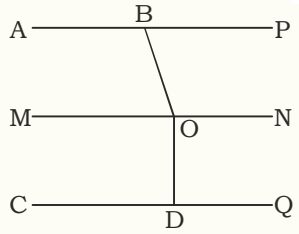
(a) $\frac{1}{2}$
(b) $\frac{1}{3}$
(c) $\frac{3}{5}$
(d) None of the above

ANSWER KEY

1.	(d)	2.	(a)	3.	(a)	4.	(b)	5.	(a)	6.	(c)	7.	(c)	8.	(c)	9.	(d)	10.	(b)
11.	(d)	12.	(a)	13.	(c)	14.	(b)	15.	(b)	16.	(d)	17.	(d)	18.	(c)	19.	(b)	20.	(c)
21.	(a)	22.	(b)	23.	(b)	24.	(c)	25.	(d)	26.	(c)	27.	(d)	28.	(b)	29.	(d)	30.	(b)
31.	(d)	32.	(a)	33.	(d)	34.	(d)	35.	(c)	36.	(a)	37.	(c)	38.	(c)	39.	(c)	40.	(c)
41.	(b)	42.	(b)	43.	(c)	44.	(b)	45.	(c)	46.	(c)	47.	(b)	48.	(c)	49.	(b)	50.	(d)
51.	(c)	52.	(a)	53.	(a)	54.	(c)	55.	(d)	56.	(a)	57.	(c)	58.	(b)	59.	(c)	60.	(d)
61.	(a)	62.	(b)	63.	(a)	64.	(c)	65.	(c)	66.	(c)	67.	(a)	68.	(c)	69.	(b)	70.	(a)

SOLUTIONS

1. (d)



$\angle ABO = 118°$

$\therefore$ MOB = $180° - 118° = 62°$

$\therefore \angle MOD = 152° - 62° = 90°$

$\therefore \angle ODC = 180° - 90° = 90°$

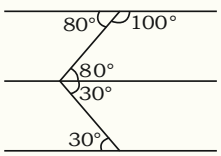
2. (a) $\angle AEP = \angle CDP = 34°$

$\angle AEP + \angle BEF + \angle DEF = 180°$

$34° + 78° + \angle DEF = 180°$

$\Rightarrow \angle DEF = 68°$

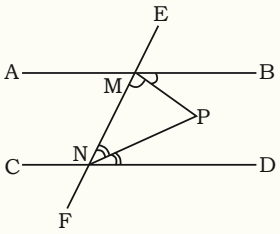
3. (a)



$\therefore x + 10° = 80° + 30° \Rightarrow x = 100°$

4. (b) Let, $\angle BMN = 2x$ and $\angle DNM = 2y$

$\therefore 2x + 2y = 180°$



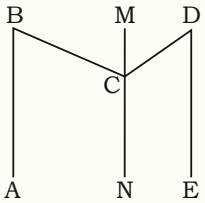
$\Rightarrow x + y = 90°$

$\therefore \angle MPN = 180° - 90° = 90°$

5. (a) $\angle BCD = \angle BCM + \angle DCM$

$= \angle ABC + \angle EDC$

$= 67° + 23° = 90°$



(Draw a line parallel to AB or DE through C)

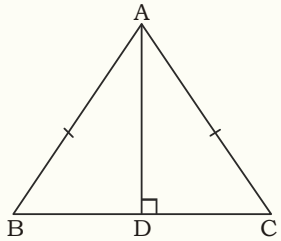
6. (c) Sides are $(x^2 - 1)$, $(x^2 + 1)$, $2x$

Let, $x = 2$

So, sides are 3, 4, 5

$\therefore$ It is right angled triangle

7. (c)



$\Rightarrow$ BD = DC = $\frac{AB}{2} = \frac{AC}{2} = \frac{BC}{2}$

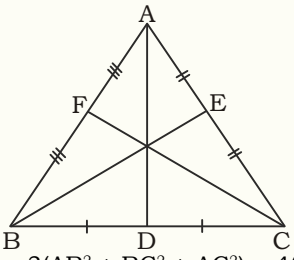
Using Pythagoras theorem,

$AD = \frac{\sqrt{3}}{2} AB = \frac{\sqrt{3}}{2} BC = \frac{\sqrt{3}}{2} CA$

$\Rightarrow AB = BC = AC = \frac{2}{\sqrt{3}} AD$

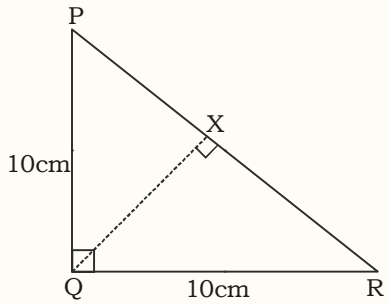
$\Rightarrow AB^2 + BC^2 + AC^2 = AD^2 \left[\frac{4}{3} + \frac{4}{3} + \frac{4}{3}\right] = 4AD^2$

8. (c)



then $3(AB^2 + BC^2 + AC^2) = 4(AD^2 + BE^2 + CF^2)$

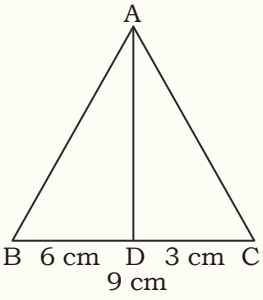
9. (d)



$PR = \sqrt{100 + 100} = 10\sqrt{2}$ cm

Then, $QX = \frac{10 \times 10}{10\sqrt{2}} = \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = 5\sqrt{2}$ cm

10. (b)



$\frac{BD}{DC} = \frac{6}{3} = \frac{2}{1}$

$\therefore \frac{\text{Area } (\Delta ABD)}{\text{Area } (\Delta ADC)} = \frac{2}{1}$

11. (d) Given,

$\angle A = 80°$, $\angle B = 40°$ and $\Delta ABC \cong \Delta FDE$

$\therefore \angle A = \angle F$, $\angle B = \angle D$ and $\angle C = \angle E$

$\because \angle C = 180° - (\angle A + \angle B)$

$\Rightarrow \angle C = 180° - 120°$

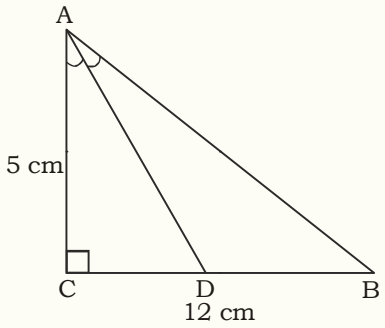
$\Rightarrow \angle C = 60°$

$\therefore \angle C = \angle E = 60°$

and AB = FD = 5 cm

Hence, Option (d) is correct.

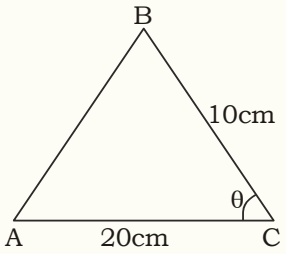
12. (a) If triangle is obtuse then [greatest angle > sum of other two angles]
So, option (a) satisfies this
$\Rightarrow 9 > 4 + 3$
$\Rightarrow 4 : 9 : 3$

13. (c)


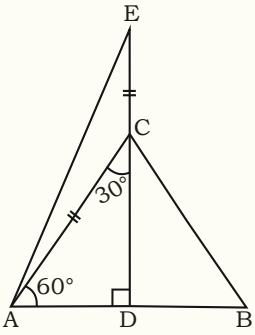
AB = 13 cm
$\because$ AD is angular bisector
$\therefore \frac{AC}{AB} = \frac{CD}{DB} = \frac{5x}{13x}$
$\because 18x = 12$ cm
$\Rightarrow x = \frac{2}{3}$ cm
$\therefore CD = 5x = 5 \times \frac{2}{3} = \frac{10}{3}$ cm
$\because AD^2 = AC^2 + CD^2 = 25 + \frac{100}{9} = \frac{225+100}{9}$
$\Rightarrow AD^2 = \frac{325}{9} \Rightarrow AD = \frac{5\sqrt{13}}{3}$ cm

14. (b) Sides = 14, 22, x
$\because$ Triangle is acute.
$\therefore x^2 + 14^2 > 22^2$
$\Rightarrow x^2 > 288$
$\because 289 = 17^2$
$\therefore x_{min} = 17$
$x_{max} = 21$
$\therefore$ Average $= \frac{17+21}{2} = \frac{38}{2} = 19$

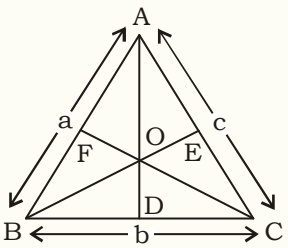
15. (b) Let, third side of triangle be 'x'.
Sides = 8, 15, x
According to property of triangle,
$15 - 8 < x < 15 + 8$
$\Rightarrow 7 < x < 23$
If a, b, c are the sides of obtuse angle triangle then,
$a^2 + b^2 < c^2$ (where, c = longest side)
If 15 is longest side then,
$8^2 + x^2 < 15^2 \Rightarrow x^2 < 161$
$\because x$ is integer
$\therefore x = 12, 11, 10, 9, 8$
If 'x' is longest side then,
$8^2 + 15^2 < x^2$
$\Rightarrow 289 < x^2 \Rightarrow 17 < x$
$\therefore x = (18, 19\ 20, 21, 22)$
$\therefore$ 10 triangles are possible.

16. (d)


Area of triangle $= \frac{1}{2} \times$ (produt of adjacent sides)$\times \sin\theta$
$\Rightarrow \frac{1}{2} \times 20 \times 10 \times \sin\theta = 80$
$\Rightarrow \sin\theta = \frac{4}{5}$
$\Rightarrow \cos\theta = \frac{3}{5}$
then, $AB^2 = AC^2 + BC^2 - 2AC \times BC \times \cos\theta$
$\Rightarrow AB^2 = (20)^2 + (10)^2 - 2 \times 20 \times 10 \times \frac{3}{5}$
$\Rightarrow AB^2 = 500 - 240 = 260$
$\Rightarrow AB = 2\sqrt{65}$ cm

17. (d)


$\Rightarrow \angle ACE = 180° - 30° = 150°$ (Linear pair)
$\because$ AC = EC (Given)
$\therefore \angle CAE = \angle CEA$
In ΔCAE,
$150° + 2\angle CAE = 180°$
$\Rightarrow \angle CAE = 15°$

18. (c)


We know, Sum of two sides of a trianlge > third side
In ΔABD,
AB + BD > AD .....(i)
In ΔACD,
AC + CD > AD .....(ii)
Adding (i) and (ii),
AB + BD + CD + AC > 2AD
$\Rightarrow$ AB + BC + AC > 2p
$\Rightarrow$ a + b + c > 2p
Similarly,

$a + b + c > 2q$

$a + b + c > 2r$

adding all the inequalities,

$3(a + b + c) > 2(p + q + r)$

$\therefore\ 2(p + q + r) < 3(a + b + c)$

19. (b) Let the centroid be O

In ΔAOB,

$AO + BO > AB$

Centroid divides median in the ratio 2 : 1

$$\therefore\ \frac{2p}{3}+\frac{2q}{3} > c$$

Similarly for other medians,

$$\frac{2q}{3}+\frac{2r}{3} > a$$

$$\frac{2p}{3}+\frac{2r}{3} > b$$

Adding the inequalities,

$$\frac{4}{3}(p+q+r) > (a+b+c)$$

$\Rightarrow 4(p + q + r) > 3(a + b + c)$

$\therefore 3(a + b + c) < 4(p + q + r)$

20. (c) (1) We know, orthocentre of obtuse angle triangle lies outside the triangle. Hence incorrect statement.

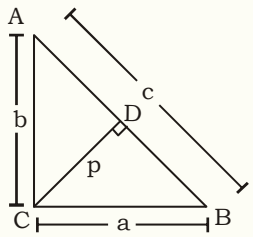
(2) Centroid always lie inside the triangle Hence true.

(3) Orthocentre of a right angle triangle lies on the vertex of triangle. Hence true.

(4) Centroid of right angle triangle lies inside the triangle. Hence incorrect statement.

Hence, Statement (2) and (3) are correct.

21. (a)



We know, area of $\Delta ABC = \frac{1}{2} \times AB \times CD$

$$= \frac{1}{2} \times c \times p \quad \ldots.(i)$$

Again, Area of $\Delta ABC = \frac{1}{2} \times BC \times AC$

$$= \frac{1}{2} \times a \times b \quad \ldots.(ii)$$

From (i) and (ii)

$$\frac{1}{2} \times c \times p = \frac{1}{2} \times a \times b$$

$$\Rightarrow cp = ab \Rightarrow c = \frac{ab}{p}$$

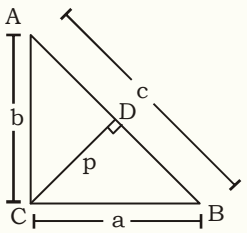
Now, by pythagorus theorem in ΔABC,

$AB^2 = AC^2 + BC^2$

$$\Rightarrow c^2 = b^2 + a^2 \Rightarrow \left(\frac{ab}{p}\right)^2 = b^2 + a^2$$

$\Rightarrow a^2b^2 = p^2(a^2 + b^2)$

Alternate Method:



By pythagorus theorem:

$$c = \sqrt{a^2 + b^2} \quad \ldots.(i)$$

We know,

$$\frac{1}{2} \times a \times b = \frac{1}{2} \times c \times p$$

$$p = \frac{AC \times BC}{AB} = \frac{b \times a}{c} \quad \ldots(ii)$$

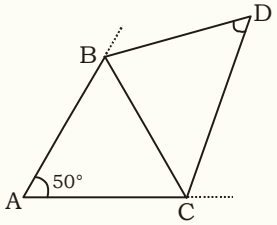
Put (i) in (ii),

$$\Rightarrow p = \frac{ab}{\sqrt{a^2 + b^2}}$$

squaring both sides,

$$p^2 = \frac{a^2b^2}{a^2 + b^2} \Rightarrow a^2b^2 = p^2(a^2 + b^2)$$

22. (b)



$\because$ Sum of angles of triangle is 180°

$\Rightarrow \angle ABC + \angle ACB = 180° - 50° \Rightarrow 130°$ ....(i)

Also, sum of angles in straight line is 180°

$$\Rightarrow \angle DBC = \frac{(180° - \angle ABC)}{2} \quad \ldots.(ii)$$

$$\Rightarrow \angle DCB = \frac{(180° - \angle ACB)}{2} \quad \ldots.(iii)$$

Adding (ii) and (iii), we get,

$$\Rightarrow \angle DBC + \angle DCB = \frac{180° - \angle ABC}{2} + \frac{180° - \angle ACB}{2}$$

$$= 90° - \frac{\angle ABC}{2} + 90° - \frac{\angle ACB}{2}$$

$$= 180° - \frac{(\angle ABC + \angle ACB)}{2}$$

Substituting value from equation (i)

$$\Rightarrow \angle DBC + \angle DCB = 180° - \frac{130°}{2}$$

$= 180° - 65° = 115°$

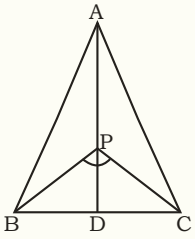
$\therefore \angle BDC = 180° - (\angle DBC + \angle DCB) = 180° - 115° = 65°$

Alternate Method:

We know, $\angle BDC = 90° - \frac{\angle A}{2}$

$$= 90° - \frac{50°}{2} = 90° - 25° = 65°$$

23. (b)



Important point: Median of a triangle divides the triangle into two parts of equal area.

In ΔABC, AD is the median of ΔABC

Area of ΔABD = Area of ΔADC .....(i)

In ΔPBC, PD is also the median of ΔPBC,

Area of ΔPBD = Area of PCD .....(ii)

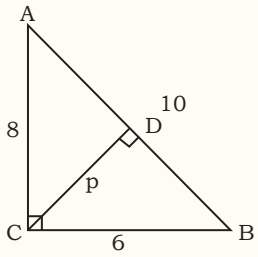
Subtracting eq. (ii) from eq. (i),

Area of ΔABD – Area of ΔPBD = Area of ΔADC – Area of ΔPCD

⇒ Area of ΔPAB = Area of ΔPAC

∴ The area of ΔPAB is equal to the area of ΔPAC

24. (c)



If ABCD is a right triangle, right angled at C and 'p' be the perpendicualr drawn from C to AB, then:

$$\frac{1}{(p)^2} = \frac{1}{(BC)^2} + \frac{1}{(CA)^2}$$

$$\Rightarrow \frac{1}{(p)^2} = \frac{1}{(6)^2} + \frac{1}{(8)^2}$$

$$\Rightarrow \frac{1}{(p)^2} = \frac{8^2+6^2}{8^2\times 6^2}$$

$$\Rightarrow p^2 = \frac{(8)^2\times(6)^2}{(10)^2}$$

$$\Rightarrow p = \frac{8\times 6}{10} = 4.8 \text{ cm}$$

Hence, the value of p is 4.8 cm.

Alternate Method:

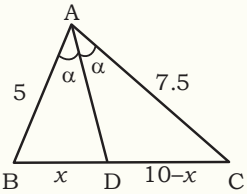
Given, AC = 8, BC = 6

⇒ AB = 10 [triplet of 6, 8, 10]

We know,

$$CD = \frac{AC\times BC}{AB} = \frac{8\times 6}{10} = 4.8 \text{ cm}$$

25. (d) In ΔABC, AD is the bisector of ∠A. If A



By angle bisector theorem

$$\frac{AB}{AC} = \frac{BD}{DC}$$

$$\Rightarrow \frac{5}{7.5} = \frac{x}{10-x}$$

$$\Rightarrow 20 - 2x = 3x$$

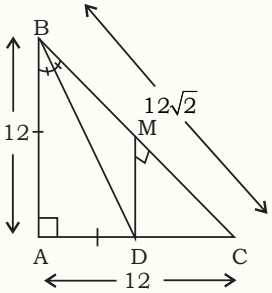
$$\Rightarrow 5x = 20 \Rightarrow x = 4$$

$$\Rightarrow BD = 4 \text{ \& } DC = 6$$

$$\text{Mid point of BC} = \frac{10}{2} = 5$$

∴ Distance of D from mid point of BC = 5 – 4 = 1 cm

26. (c)



By Internal angle bisector theorem,

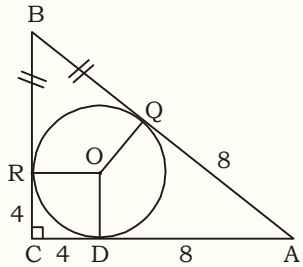
$$\frac{AB}{BC} = \frac{AD}{DC} \Rightarrow \frac{12}{12\sqrt{2}} = \frac{AD}{DC}$$

$$\Rightarrow \frac{1}{\sqrt{2}} = \frac{AD}{DC}$$

$(\sqrt{2} + 1)$ units = 12

$$1 \text{ unit} = \frac{12}{\sqrt{2}+1} = 12\left(\sqrt{2}-1\right) \text{ cm}$$

27. (d)



Let, BR = x

In ΔBCA, $AB^2 = AC^2 + BC^2$

$\Rightarrow (8 + x)^2 = 12^2 + (4 + x)^2$

$\Rightarrow 64 + x^2 + 16x = 144 + 16 + x^2 + 8x$

$\Rightarrow 8x = 160 - 64$

$\Rightarrow 8x = 96$

$\Rightarrow x = 12$ cm

∴ CB = 12 + 4 = 16 cm

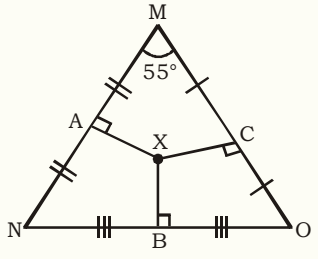
SMART APPROACH:-

With the help of triplets,

3	4	5
↓	↓	↓
12	16	20

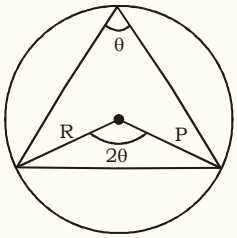
CB = 16 cm because it satisfied the given conditions

28. (b)



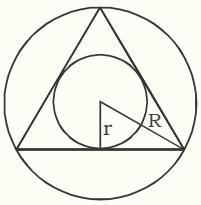
$\angle NMO = 55°$ (given)

We know that,



$\therefore \angle NXO = 55° \times 2 = 110°$

29. (d)



Length of inradius (r) $= \frac{a}{2\sqrt{3}} = \frac{18}{2\sqrt{3}} = 3\sqrt{3}$

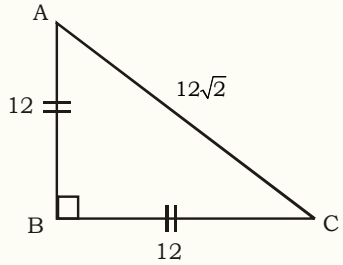
Length of circumradius (R) $= \frac{a}{\sqrt{3}} = 6\sqrt{3}$

ATQ,

Area of region between circle $= \pi (R^2 - r^2)$

$= \frac{22}{7}(108 - 27) = \frac{22}{7} \times 81 = 254\frac{4}{7}$

30. (b)



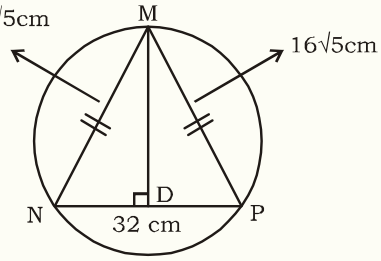
In radius $= \frac{P + B - H}{2}$

$= \frac{12 + 12 - 12\sqrt{2}}{2} = 12 - 6\sqrt{2} = 6(2 - \sqrt{2})$

Circum radius $= \frac{H}{2} = \frac{12\sqrt{2}}{2} = 6\sqrt{2}$

$\therefore$ Required ratio $= \frac{6(2 - \sqrt{2})}{6\sqrt{2}} = \frac{2 - \sqrt{2}}{\sqrt{2}}$

31. (d)



MD is perpendicular to NP

So, ND = DP $= \frac{32}{2} = 16$ cm

$MD = \sqrt{\left[(16\sqrt{5})^2 - (16)^2\right]}$

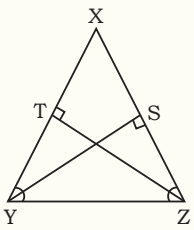
$= \sqrt{(1280 - 256)} = \sqrt{1024} = 32$ cm

Area of $\Delta MNP = \frac{1}{2} \times 32 \times 32 = 512$ cm^2

$\therefore$ Circumradius $= \frac{abc}{4 \times \text{Area of triangle}}$

$= \frac{16\sqrt{5} \times 16\sqrt{5} \times 32}{4 \times 512} = 20$cm

32. (a)



$\because$ XY = XZ

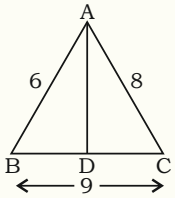
$\Rightarrow$ Area of ΔXYZ = Area of ΔXZY

$\Rightarrow \frac{1}{2} \times XY \times ZT = \frac{1}{2} \times XZ \times YS$

$\therefore$ XY = XZ, Then

$\Rightarrow \frac{1}{2} \times XY \times ZT = \frac{1}{2} \times XY \times YS \Rightarrow ZT = YS$

33. (d)



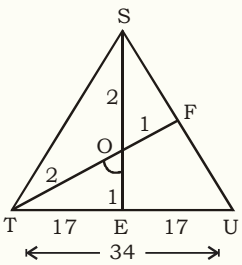
By Apollonius theorem,

$(AB)^2 + (AC)^2 = 2\left[(AD)^2 + \left(\frac{BC}{2}\right)^2\right]$

$\Rightarrow (6)^2 + (8)^2 = 2\left[(AD)^2 + \left(\frac{9}{2}\right)^2\right]$

$\Rightarrow AD = \frac{\sqrt{119}}{2}$ cm

34. (d)



If two medians intersect at a point that point called centroid (O).

SO : OE = 2 : 1

$OE = 24 \times \frac{1}{3} = 8$ cm

TO : OF = 2 : 1

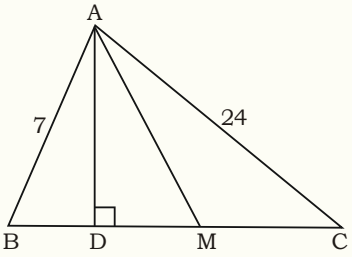
$TO = 22.5 \times \frac{2}{3} = 15$ cm

So, triangle ΔTOE is a right angle triangle.

$\therefore \angle TOE = 90°$

35. (c) $\because$ AM is the median of a right angled triangle.

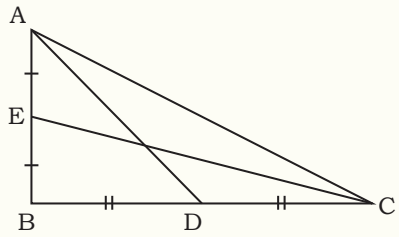
$\therefore AM = \frac{BC}{2} = \frac{25}{2}$



$AD = \frac{AB \times AC}{BC}$

$\frac{AD}{AM} = \frac{7 \times 24 \times 2}{25 \times 25} = \frac{336}{625}$

36. (a)



AE = BE and BD = CD

$AB^2 = AC^2 - BC^2 = 25 - BC^2$

and $AB^2 = AD^2 - BD^2 = \left(\frac{3\sqrt{5}}{2}\right)^2 - BD^2 = \frac{45}{4} - \frac{BC^2}{4}$

From eq. (i) and (ii)

$BC^2 = \frac{55}{3}$

Now, from eq. (i)

$AB^2 = 25 - \frac{55}{3} = \frac{20}{3}$

Also $CE^2 = BE^2 + BC^2$

$= \left(\frac{1}{2}AB\right)^2 + BC^2 = \frac{AB^2}{4} + BC^2 = CE^2$

$= \frac{5}{3} + \frac{55}{3} = \frac{60}{3} = 20$

$CE = 2\sqrt{5}$ cm

37. (c) $\angle EAB = \angle OAB = 60°$

$\angle ABD = 45°$

$\angle AOB = 180° - (60° + 45°) = 75°$

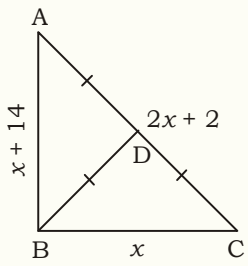
38. (c) Let, ΔABC is a Right angle triangle & BD is median

In rightangle triangle, BD = AD = DC

ATQ,

BC = x, hypotenuse (AC) = $2x + 2$

and AB = $\frac{2x+2}{2} + 13 = x + 14$



From Pythagorean theorem,

$(2x + 2)^2 = x^2 + (x + 14)^2$

$\Rightarrow 4x^2 + 4 + 8x = x^2 + x^2 + 196 + 28x$

$\Rightarrow 2x^2 - 20x - 192 = 0$

$\Rightarrow x^2 - 10x - 96 = 0$

$\Rightarrow (x + 6)(x - 16) = 0$

$x = 16, -6$

$\therefore$ BD = $\frac{AC}{2} = \frac{2 \times 16 + 2}{2} = 17$

Alternate Method:-

Let, ΔABC is a Right angle triangle & BD is median

In rightangle triangle, BD = AD = DC

ATQ,

BC = x, hypotenuse (AC) = $2x + 2$

and AB = $\frac{2x+2}{2} + 13 = x + 14$

16, 30 and 34 follows the above condition.

$\therefore$ BD = $\frac{AC}{2} = \frac{2 \times 16 + 2}{2} = 17$

39. (c) s = $\frac{3+6+5}{2} = 7$

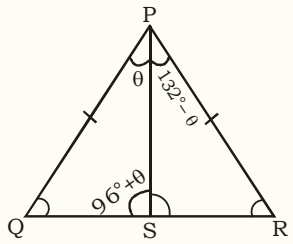
Area of triangle formed by medians

$= \frac{4}{3}\sqrt{s(s-a)(s-b)(s-c)}$

$= \frac{4}{3}\sqrt{7(7-3)(7-6)(7-5)}$

$= \frac{4}{3}\sqrt{7 \times 4 \times 1 \times 2} = \frac{4}{3}\sqrt{56}$

40. (c)



$\angle PQR = \angle PRQ$ ($\because$ PQ = PR)

$= \frac{180° - 132°}{2} = \frac{48°}{2} = 24°$

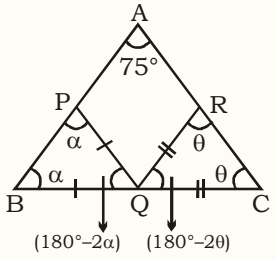
In ΔPQS, $24° + \theta + 96° + \theta = 180°$

$\Rightarrow 2\theta = 180° - 120° \Rightarrow 2\theta = 60°$

$\Rightarrow \theta = 30°$

$\therefore \angle PSR = 180° - (96° + 30°) = 54°$

41. (b)



In ΔABC, $\alpha + \theta + 75° = 180°$

$\Rightarrow \alpha + \theta = 105°$

$\because$ BQC is a straight line

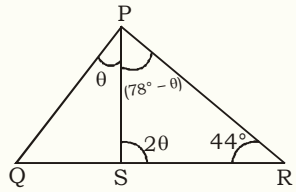
$180° - 2\alpha + \angle PQR + 180° - 2\theta = 180°$

$\Rightarrow \angle PQR = 180° - 360 + 2(\alpha + \theta)$

$= -180° + 2 \times 105°$
$= -180° + 210° = 30°$

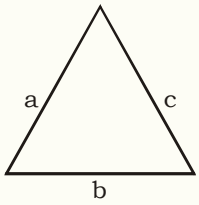
SMART APPROACH:-
$\angle PQR = 180° - 2 \times \angle BAC$
$= 180° - 2 \times 75° = 30°$

42. (b)



In ΔPSR, $78° - \theta + 2\theta + 44° = 180°$
$\Rightarrow \theta = 58°$ (angle sum property)
$\angle PSQ = 180° - 2\theta = 180° - 2 \times 58°$
(Straight angle) $= 180° - 116° = 64°$

43. (c)

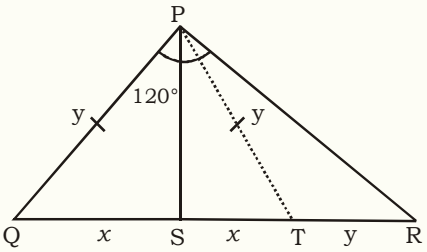


$\Rightarrow a + b > c$
The sum of two sides is always greater than the third side.

44. (b) $12 + 7 > x > 12 - 7$
$19 > x > 5$
$x = (19 - 5) - 1 = 13$

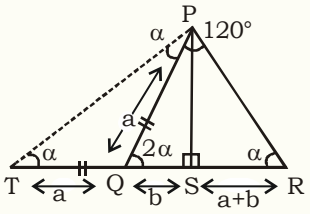
SMART APPROACH:-
$2 \times$ smallest side $- 1$
$= 2 \times 7 - 1 = 13$

45. (c)



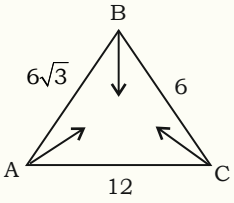
Given,
$\angle P = 120°$, $PS \perp QR$ and $PQ + QS = SR$.
$PT = PQ$. Let $PQ = y$ and $QT = 2x$, $\angle PQS = \theta$ and $\angle PRQ = \alpha$
In ΔPQR, $\angle PQR + \angle PRQ + \angle QPR = 180°$
$\Rightarrow \theta + \alpha = 180° - 120° = 60°$ ...(i)
$\Rightarrow$ PQT is an isosceles triangle and $PS \perp QR$
$\Rightarrow QS = ST = x$
$\Rightarrow \angle PTQ = \theta$..... (PQ = PT)
And $\angle QPT = 180° - 2\theta$
Now, $PQ + QS = SR$ ..... (Given)
$\Rightarrow y + x = x + TR$
$\Rightarrow TR = y$
$\Rightarrow PT = TR = y$
$\Rightarrow \angle PRT = \angle TPR = \alpha$ ....(PT = TR)
ATQ,
$\angle QPT + \angle TPR = 120°$
$\Rightarrow 180° - 2\theta + \alpha = 120°$
$\Rightarrow 2\theta - \alpha = 60°$ .....(ii)
Add eq. (i) and (ii)
$\Rightarrow \theta = 40°$

Alternate Method:



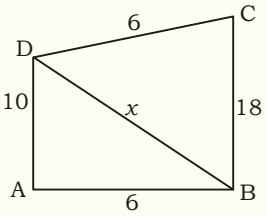
Extend RQ to T such that PQ = TQ,
Let, $\angle PTQ = \alpha$,
$\Rightarrow \angle TPQ = \alpha$ (TQ = PQ)
$\therefore \angle PQS = 2\alpha$ (External angle property)
$\Delta PSR \cong \Delta PST$
$\Rightarrow PS = PS$
$SR = TS = a + b$
$\angle PSR = \angle PSQ = 90°$
$\Rightarrow \angle PRS = \angle PTS = \alpha$
In ΔPTR, $\alpha + \alpha + \alpha + 120° = 180°$
$\Rightarrow 3\alpha = 60° \Rightarrow \alpha = 20°$
$\therefore \angle Q = 2\alpha = 2 \times 20° = 40°$

46. (c)



$\therefore$ BC AB AC
$\Rightarrow$ 6 : $6\sqrt{3}$: 12
$\Rightarrow$ 1 : $\sqrt{3}$: 2
30° 60° 90°
$\therefore \angle B = 90°$

47. (b)



In ΔDAB:-
We know that,
$10 - 6 < x < 10 + 6$
$4 < x < 16$ ....(i)
In ΔDCB :-
$18 - 6 < x < 18 + 6$
$12 < x < 24$ .....(ii)
In eqn (i) & (ii)
$12 < x < 16$

48. (c) Given triangle is right angled triangle.
Triplet with minimum integer value is (3,4,5).
Now,
Smallest side $\Rightarrow 3x = 3 \Rightarrow x = 1$
Middle side $\Rightarrow 4\sqrt{y} = 4 \Rightarrow \sqrt{y} = 1 \Rightarrow y = 1$
Largest side $\Rightarrow 5\sqrt[3]{z} = 5$
$\Rightarrow \sqrt[3]{z} = 1 \Rightarrow z = 1$
$\therefore$ The minimum values of x, y and z are 1, 1, 1.

49. (b) We have, $2\angle A = 3\angle B = 6\angle C$
$\Rightarrow 2\angle A = 6\angle C$ & $3\angle B = 6\angle C$
$\Rightarrow \angle A = 3\angle C \Rightarrow \angle B = 2\angle C$
Now,
$\Rightarrow \angle A + \angle B + \angle C = 180°$ (sum of interior angles of a triangle)
$\Rightarrow 3\angle C + 2\angle C + \angle C = 180°$
$\Rightarrow 6\angle C = 180°$
$\Rightarrow \angle C = 30°$
$\angle A = 3\angle C = 3 \times 30 = 90°$
$\therefore \angle A + \angle C = 90° + 30° = 120°$

Alternate Method:

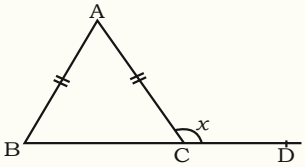
$2\angle A = 3\angle B = 6\angle C = k$ (let)
$\angle A = \angle B = \angle C = 3k : 2k : k$
$\Rightarrow \angle A + \angle B + \angle C = 180°$ (sum of interior angles of a triangle)
$\Rightarrow 3k + 2k + k = 180°$
$\Rightarrow k = 30°$
$\angle A = 3k = 3 \times 30° = 90°$
$\therefore \angle A + \angle C = 90° + 30° = 120°$

50. (d) Let the sides of right angle triangle are p, p + 2 and p + 4
As in right angled triangle biggest side is hypotenuse
$\Rightarrow (p + 4)^2 = (p)^2 + (p + 2)^2$
$\Rightarrow p^2 + 16 + 8p = p^2 + p^2 + 4p + 4$
$\Rightarrow p^2 - 4p - 12 = 0$
$\Rightarrow (p - 6)(p + 2) = 0$
$\Rightarrow p = 6$ or $p = -2$ (reject as the side is always positive)
$\therefore$ The sides of triangle are (p = 6) , (p + 2 = 8) , (p + 4 = 10)
$\therefore$ Product of these integers = $6 \times 8 \times 10 = 480$

Alternate Method:

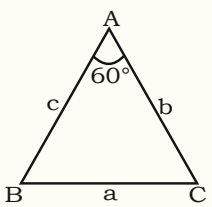
We know, (6, 8, 10) is a triplet and it has even number which are consecutive
$\therefore$ Product = $6 \times 8 \times 10 = 480$

51. (c) Since, ΔABC in an isosceles triangle
So,



$\angle ABC = \angle ACB$ [$\because$ AB = AC) .....(i)
According to the figure,
$\angle ACB + \angle ACD = 180°$ (Linear pair angles)
$\Rightarrow \angle ACB + x = 180°$
$\Rightarrow \angle ACB = (180° - x)$ ....(ii)
From (i) & (ii), We have:-
$\Rightarrow \angle ABC = (180° - x)$ ...(iii)
Now,
In ΔABC,
$\Rightarrow \angle ABC + \angle ACB + \angle BAC = 180°$ (Sum of all angles of a triangle)
$\Rightarrow 180° - x + 180° - x + \angle BAC = 180°$
$\Rightarrow \angle BAC = 180° - 360° + 2x$
$\Rightarrow \angle BAC = 2x - 180°$
$\therefore$ The value of $\angle BAC$ is $2x - 180°$

52. (a)



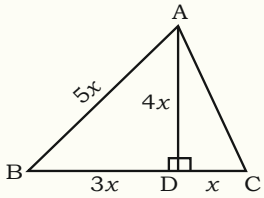
By cosine rule:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \cos 60° = \frac{(AB)^2 + (AC)^2 - (BC)^2}{2(AB \times AC)}$$

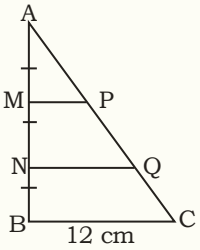
$\Rightarrow AB \times AC = AB^2 + AC^2 - BC^2$
$\therefore AB^2 + AC^2 - BC^2 = AB.AC$

53. (a)



Let CD = x, DB = $3x$, AD = $4x$ & AB = $5x$
In ΔADC,
$(AC)^2 = (AD)^2 + (CD)^2$
$\Rightarrow AC = \sqrt{16x^2 + x^2}$
$\Rightarrow AC = \sqrt{17}x$
Now,
$(AB + AC)(AB - AC) = (AB)^2 - (AC)^2 = 25x^2 - 17x^2 = 8x^2$
From option (a)
$2(AB + AC)(AB - AC) = (BC)^2$
$\Rightarrow 2(8x^2) = (3x + x)^2$
$\Rightarrow 16x^2 = 16x^2$
Hence Option (a) is correct.

54. (c) Here, we have:
AM = MN = NB



In ΔAMP and ΔABC,
$\angle AMP = \angle ABC = 90°$
MP || BC
So, ΔAMP ~ ΔABC

$$\Rightarrow \frac{AM}{AB} = \frac{MP}{BC} \Rightarrow \frac{1}{3} = \frac{MP}{12}$$

MP = 4 cm
Similarly, ΔANQ and ΔABC are similar triangle as $\angle ANQ = \angle ABC = 90°$ and NQ || BC

$$\text{So, } \frac{AN}{AB} = \frac{NQ}{BC}$$

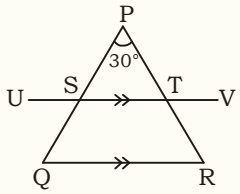
$$\Rightarrow \frac{2}{3} = \frac{NQ}{12} \Rightarrow NQ = 8 \text{ cm}$$

MP + NQ = 4 + 8 = 12 cm

55. (d) XY is parallel to BC and remaining two sides are common.
ΔABC and ΔAXY are similar triangles, i.e. ΔAXY is also an equilateral triangle.
$\Rightarrow XY = AX = AY$
Similarly, ΔBXP and ΔQYC are also similar to ΔABC and are equilateral triangles.
$\Rightarrow XP = BX = BP$
$\Rightarrow YQ = YC = QC$

Now,
$\Rightarrow$ AB = AX + XB = XY + XP .....(i)
$\Rightarrow$ AC = AY + YC = XY + YQ ..... (ii)
$\Rightarrow$ BC = BP + PQ + QC
$\Rightarrow$ BC = XP + PQ + YQ ..... (iii)
Adding equation (i), (ii) and (iii)
$\Rightarrow$ AB + AC + BC = 2 (XY + XP + YQ) + PQ
Substituting values,
$\Rightarrow$ 30 + 30 + 30 = 2 × 40 + PQ
$\Rightarrow$ PQ = 90 – 80 = 10 cm

56. (a)

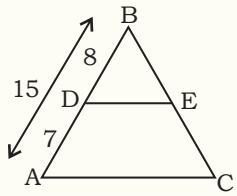


1. $\frac{PS}{PQ} = \frac{PT}{PR}$ (Thales theorem)

2. $ST \neq \frac{QR}{2}$ (S and T are not midpoints)

3. $\Delta PST \sim \Delta PQR$
$\angle P = \angle P$ (Common angle)
$\angle S = \angle Q$
and $\angle T = \angle R$ (corresponding angles)
Hence, statement (1) and (3) are true

57. (c)



ATQ,

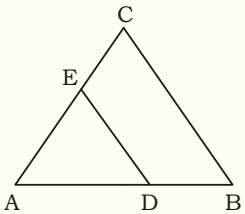
$$\frac{\text{Area of } \Delta BDE}{\text{Area of } \Delta BAC} = \frac{BD^2}{AB^2}$$

$$\Rightarrow \frac{\text{Area of } \Delta BDE}{\text{Area of } \Delta BAC} = \frac{8^2}{15^2}$$

$$\Rightarrow \frac{\text{Area of } \Delta BDE}{\text{Area of } \Delta BAC} = \frac{64}{225}$$

Area of trapezium = 225 – 64 = 161
$\therefore$ Required ratio = 64 : 161

58. (b) In the following diagram:



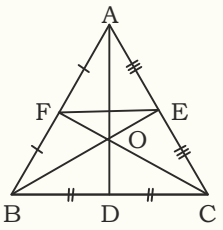
$\Delta AED \sim \Delta ACB$

$$\Rightarrow \frac{AD}{AB} = \frac{AE}{AC} \Rightarrow \frac{x}{x+x-2} = \frac{x+2}{x+2+x-1}$$

$$\Rightarrow \frac{x}{2x-2} = \frac{x+2}{2x+1}$$

$\Rightarrow 2x^2 + 4x - 2x - 4 = 2x^2 + x$
$\Rightarrow x = 4$

59. (c)

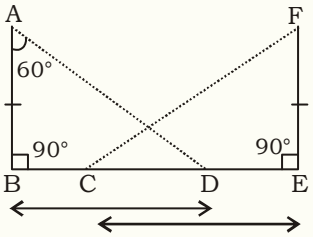


$$\frac{1}{4} \times \text{Area of } \Delta ABC = \text{Area of } \Delta AFE$$

$$\Rightarrow \text{Area of } \Delta AFE = \frac{1}{4} \times 20 = 5 \text{ cm}^2$$

Area of trapezium FBCE = Area of ΔABC – Area of ΔAFE
= (20 – 5) = 15 cm^2

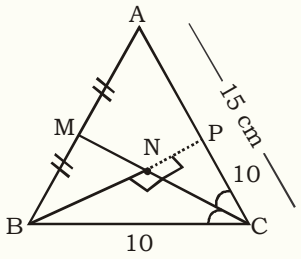
60. (d)



In ΔABD and ΔFEC,
AB = FE (given)
BD = CE (given)
$\angle BAD = 60°$ (given)
$\angle B = \angle E = 90°$
$\Delta ABD \cong \Delta FEC$ (from SAS)
$\Rightarrow \angle F = 60°$
$\therefore \angle FCE = 180° - 90° - 60° = 30°$

$$\therefore \frac{\angle BAD}{\angle FCE} = \frac{60°}{30°} = \frac{2}{1} = 2 : 1$$

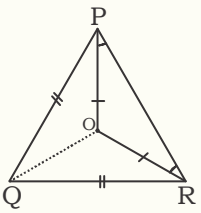
61. (a)



In ΔCBN and ΔCPN,
NC = NC (common)
$\angle BCN = \angle PCN$ (Given)
$\angle CNB = \angle CNP$ (each 90°)
$\Rightarrow$ CBN $\cong \Delta CPN$ (By ASA)
$\Rightarrow$ NP = NB (i.e N is the midpoint of BP)
CP = CB = 10 cm
AP = 15 – 10 = 5 cm

In ΔABP, $MN = \frac{1}{2} AP = \frac{1}{2} \times 5 = 2.5$ cm

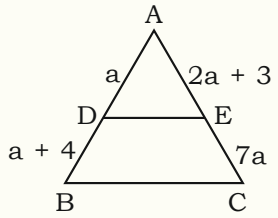
62. (b)



$\therefore$ PQ = QR
Then ΔPQR is a isosceles Δ.
Statement (I) is true

Intersection point of median is called centeroid. Hence statement (II) is wrong
In ΔPOQ and ΔQOR
PQ = QR (Given)
PO = OR ($\therefore \angle OPR = \angle ORP$)
OQ = OQ
$\Delta POQ \cong \Delta QOR$
Statement (III) is true.
Hence, option (b) is right answer.

63. (a)



$\because DE \parallel BC$
$\Delta ADE \sim \Delta ABC$
By Thales Law,

$$\frac{AD}{AB} = \frac{AE}{AC}$$

$$\Rightarrow \frac{a}{a+a+4} = \frac{2a+3}{2a+3+7a}$$

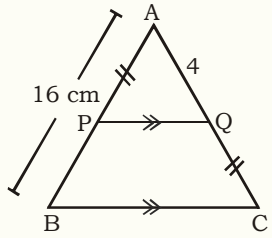
$$\Rightarrow \frac{a}{2a+4} = \frac{2a+3}{9a+3}$$

$\Rightarrow 9a^2 + 3a = 4a^2 + 6a + 8a + 12$
$\Rightarrow 5a^2 - 11a - 12 = 0$
$\Rightarrow 5a^2 - 15a + 4a - 12 = 0$
$\Rightarrow 5a(a-3) + 4(a-3) = 0$
$\Rightarrow (a-3)(5a+4) = 0$
$\Rightarrow a - 3 = 0 \Rightarrow a = 3$

$5a + 4 = 0 \Rightarrow a = -\frac{4}{5}$ (Not possible)

$\therefore \quad a = 3$

64. (c)



Let, QC = x = AP
Then, By Thales Law,

$$\frac{AP}{AB} = \frac{AQ}{AC}$$

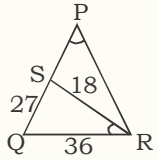
$$\Rightarrow \quad \frac{x}{16} = \frac{4}{4+x}$$

$\Rightarrow \quad x^2 + 4x - 64 = 0$
By quadratic formula,

$$x = \frac{-4 \pm \sqrt{16+256}}{2}$$

$$\Rightarrow x = \frac{\sqrt{272}-4}{2} = \frac{4\sqrt{17}-4}{2} = 2\sqrt{17}-2$$

65. (c)



In ΔQPR and ΔQRS,
$\angle QPR = \angle QRS$
$\angle Q = \angle Q$ (Common)
QR = QR = (Common side)
$\Delta QPR \sim \Delta QRS$

$$\Rightarrow \frac{PQ}{QR} = \frac{PR}{RS} = \frac{QR}{QS}$$

$$\Rightarrow \frac{PQ}{36} = \frac{PR}{18} = \frac{36}{27}$$

$$\Rightarrow PQ = \frac{36\times36}{27} = 48 \text{ cm}$$

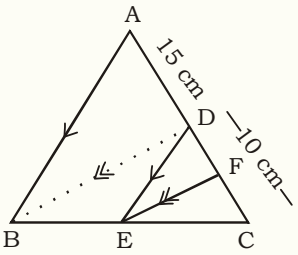
$$PR = \frac{36\times18}{27} = 24$$

$\therefore PS = 48 - 27 = 21$ cm

$$\therefore \frac{\text{Perimeter of } \Delta PRS}{\text{Perimeter of } \Delta QSR} = \frac{24+18+21}{27+36+18} = \frac{63}{81} = \frac{7}{9}$$

SMART APPROACH:-
Perimeter of triangle QSR
= 27 + 36 + 18 = 81 cm
So, in required ratio, denominator must be factor of 81.
In option (c), 9 is the factor of 81
So, option (c) is correct.

66. (c)



$\Delta CEF \sim \Delta CBD \qquad (\because EF \parallel BD)$

$$\Rightarrow \frac{CE}{BC} = \frac{CF}{CD} \quad \ldots.(i)$$

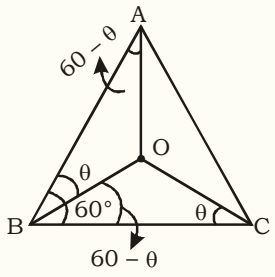
Also, $\Delta CED \sim \Delta CBA \quad (\because DE \parallel AB)$

$$\Rightarrow \frac{CE}{BC} = \frac{CD}{CA} \quad \ldots.(ii)$$

From (i) and (ii)

$$\frac{CF}{CD} = \frac{CD}{CA} \Rightarrow CF = \frac{CD^2}{CA} = \frac{100}{25} = 4$$

67. (a)



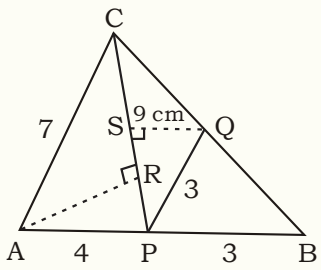
$$\angle AOB = \angle BOC = \angle COA = \frac{360^\circ}{3} = 120^\circ$$

In ΔAOB and ΔBOC
$\angle OAB = \angle OBC \Rightarrow \angle OBA = \angle OCB$
$\angle O = \angle O$
$\Delta AOB \sim \Delta BOC$ (by AAA)

$$\Rightarrow \frac{AO}{BO} = \frac{OB}{OC} \Rightarrow \frac{12}{BO} = \frac{BO}{9}$$

$\Rightarrow BO^2 = 108 \Rightarrow BO = 6\sqrt{3}$ cm

68. (c)



In ΔACR and ΔQPS

∠ARC = ∠PSQ (each 90°)

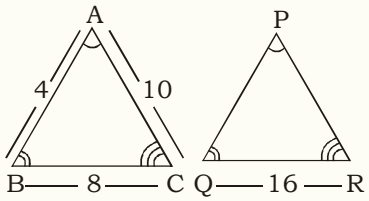
∠ACP = ∠SPQ (Alternate angle)

So, ΔACR ~ ΔQPS

$\Rightarrow \frac{PQ}{AC} = \frac{SQ}{AR} \Rightarrow \frac{3}{7} = \frac{9}{AR}$

$\Rightarrow AR = \frac{9 \times 7}{3} = 21$ cm

69. (b)



In ΔABC,

Semi–perimeter of ΔABC (s) = $\frac{4+8+10}{2} = 11$

Area = $\sqrt{s(s-a)(s-b)(s-c)}$

= $\sqrt{11 \times 3 \times 1 \times 7} = \sqrt{231}$ cm²

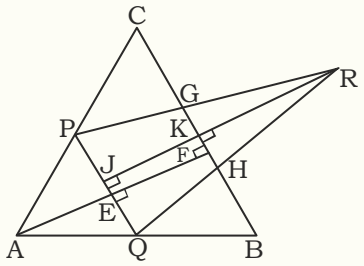
ΔABC ~ ΔPQR

$\frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta PQR} = \frac{BC^2}{QR^2} = \frac{8^2}{16^2} = \frac{1}{4}$

1 unit → $\sqrt{231}$ cm²

4 unit → $4\sqrt{231}$ cm²

70. (a)



ΔAPQ ~ ΔACB , BC = 2PQ

and BC | | PQ

⇒ AE = 2AF

⇒ AE = EF

Again ΔRGH ~ ΔRPQ

and PQ = 2GH (By mid-point theorem)

⇒ RJ = 2RK

⇒ RK = JK

But since EF = JK

∴ AE = EF = JK = RK

∴ RJ = RK + JK and AF = AE + EF

and RJ = AF = h (say),

then, $\frac{\text{Area of } \Delta PQR}{\text{Area of } \Delta ABC} = \frac{\frac{1}{2} \times PQ \times h}{\frac{1}{2} \times BC \times h} = \frac{PQ}{BC} = \frac{1}{2}$